

Solution Architecture

Filled project document for GitHub and SkillWallet submission

Date	28 June 2026	Team ID	To be updated in SkillWallet
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction	Team Members	Devireddy Prasanna, Chokka Abitha, Navya Borugadda, Bodi Varshitha, Mulakala Harika Venkata Sridevi

Architecture Layers

Layer	Responsibility	Artifacts
User Layer	User accesses app from web browser.	Browser
Presentation Layer	Dashboard, input form, prediction result, optional history.	HTML, CSS, JavaScript
Application Layer	Route handling, validation, scaling, model loading, prediction.	Flask app.py
Machine Learning Layer	Trained model and preprocessing object.	model.pkl, scaler.pkl
Data Layer	Dataset, preprocessing, feature engineering, train/test split.	CSV, notebook
Deployment Layer	Git push, GitHub repository, IBM Cloud server, live application.	GitHub, IBM Cloud

Data Flow

User enters weather readings -> Flask validates input -> scaler transforms values -> saved ML model predicts risk -> application returns Flood or No Flood -> result is shown to the user and may be used for warning or resource planning.